

ASTRODYNAMICS I: HOMEWORK #5

Due: Mon, June 4, 11:00AM, 2018

1. Halley's comet has a period of 76 years and its closest approach to the center of the Sun is 90 million km. The Sun's gravitational parameter is $\mu_S = 1.327 \times 10^{11} \text{ km}^3/\text{sec}^2$.
 - a. What is the specific angular momentum of the orbit of Halley's comet? What is the specific energy of the orbit of Halley's comet?
 - b. What is the orbital eccentricity of Halley's comet?
 - c. What are the radius of aphelion and of perihelion of the orbit of Halley's comet?
 - d. What is the orbital velocity vector of Halley's comet at perihelion; what is the orbital velocity vector of Halley's comet at aphelion?
2. Example 4-1 in Chapter 4 of Vallado (pp. 260-261).